

Progress Report

Team One Parallel

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Summary

- **Long-Term Goal:** Monte Carlo simulation of high-NA optical pulses with extended DOF (annular apertures) in scattering media
- **Core Problem:**
 - Imaging through scattering media requires both *tight focusing* (high NA) and *scattering simulation* (Monte Carlo)
 - Current Monte Carlo implementations use scalar diffraction (breaks down at high NA)
 - Optical pulses ($\tau < 2$ fs) exhibit spectral broadening that further complicates focusing
- **Gap**
 - No existing tool combines high-NA vector diffraction + pulsed sources + Monte Carlo scattering

Summary

- **Current Work** (this report)
 - Build and validate Richards-Wolf vector diffraction tools
 - Extend to annular apertures for DOF control
 - Quantify when scalar approximation fails
- **Next Steps:**
 - Add pulsed source support (spectral integration)
 - Later: Integrate with Monte Carlo photon tracing for scattering media

Reference Papers

PHYSICAL REVIEW A **68**, 033812 (2003)

Three-dimensional light distribution near the focus of a tightly focused beam of few-cycle optical pulses

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Via the Richards-Wolf vector diffraction theory, we analyze the three-dimensional intensity distribution of the focal volume that is produced by a strongly focused 750-nm beam of ultrafast, Gaussian-shaped optical pulses (10^{-9} s $\gg$ pulse width $\tau \gg 1$ fs = 10^{-15} s). Knowledge of the three-dimensional distribution near focus is essential in determining the diffraction-limited resolution of an optical microscope. The optical spectrum of a short pulse is characterized by side frequencies about the carrier frequency. The effect of spectral broadening on the focused intensity distribution is evaluated via the Linfoot's criteria of fidelity, structural content, and correlation quality and with reference to a 750-nm cw focused beam. Different values are considered for τ and numerical aperture of the focusing lens ($0.1 \leq X_{\text{NA}} \leq 1.2$). At $X_{\text{NA}} = 0.8$, rapid deterioration of the focused intensity distribution is observed at $\tau = 1.2$ fs. This happens because a 750-nm optical pulse with $\tau = 1.2$ fs has an associated coherence length of 359.7 nm which is less than the Nyquist sampling interval of 375 nm that is required to sample 750 nm sinusoid without loss of information. The ill-effects of spectral broadening is weaker in two-photon excitation microscope than in its single-photon counterpart for the same focusing lens and light source.

DOI: 10.1103/PhysRevA.68.033812

PACS number(s): 42.65.Re, 87.64.Rr, 87.64.Tt, 87.64.Vv

Analyzes how few-cycle optical pulses deteriorate focal quality due to spectral broadening effects.

Monte Carlo analysis of focused Gaussian beam in scattering media: Curvature correction and Mie scattering

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ARTICLE INFO

Dataset link: <https://github.com/harjonillo/gb-p-mc>

Keywords:

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Light scattering
Focused beam propagation in scattering media

ABSTRACT

We introduce curvature correction into the Monte Carlo (MC) technique to determine the suitable photon-step size and initial photon distribution in the focusing lens aperture, that can account for the effects of the radius of curvature on the axial intensity profile of a focused Gaussian beam that is propagated through a scattering medium. The scattering anisotropy of the medium is determined via the Mie scattering theory for given values of the scattering particle radius, density, and phase distribution values while optical ray tracing is used to determine the occurrence of scattering events. We evaluate the performance of the modified MC technique for random and periodic media by examining the axial intensity profiles of the propagating focused beam. Relative to the predictions of the scalar diffraction theory, curvature correction improves the accuracy of the MC results with decreasing step size s and increasing number of steps N_s . In the absence of scattering, the axial beam distribution approaches that of the scalar diffraction theory for a given numerical aperture ($NA \leq 0.5$), s , and N_s based on the Linfoot's image quality criteria of fidelity, correlation quality and structural content.

Introduces curvature correction to improve Monte Carlo accuracy for focused beam propagation in scattering media.

Research Context

Key Applications

- Mouse embryo imaging (thick biological samples)
- Integrated circuit inspection
- Attosecond x-ray generation

Current Approaches

- **Scalar Diffraction Theory (SDT)**
 - Reliable for $NA \leq 0.5$
 - Computationally efficient but limited scope
 - Scalar approximation, no polarization effects

Current Approaches

- **Richards-Wolf Vector Diffraction Theory**
 - Handles tightly focused beams ($\text{NA} > 0.5$)
 - Full vector field calculations
 - Accounts for polarization effects
 - Assumes monochromatic illumination

Vector field components:

$$E_x(P) = -j(I_0 - I_2 \cos \phi)$$

$$E_y(P) = -jI_2 \sin \phi$$

$$E_z(P) = -2I_1 \cos \phi$$

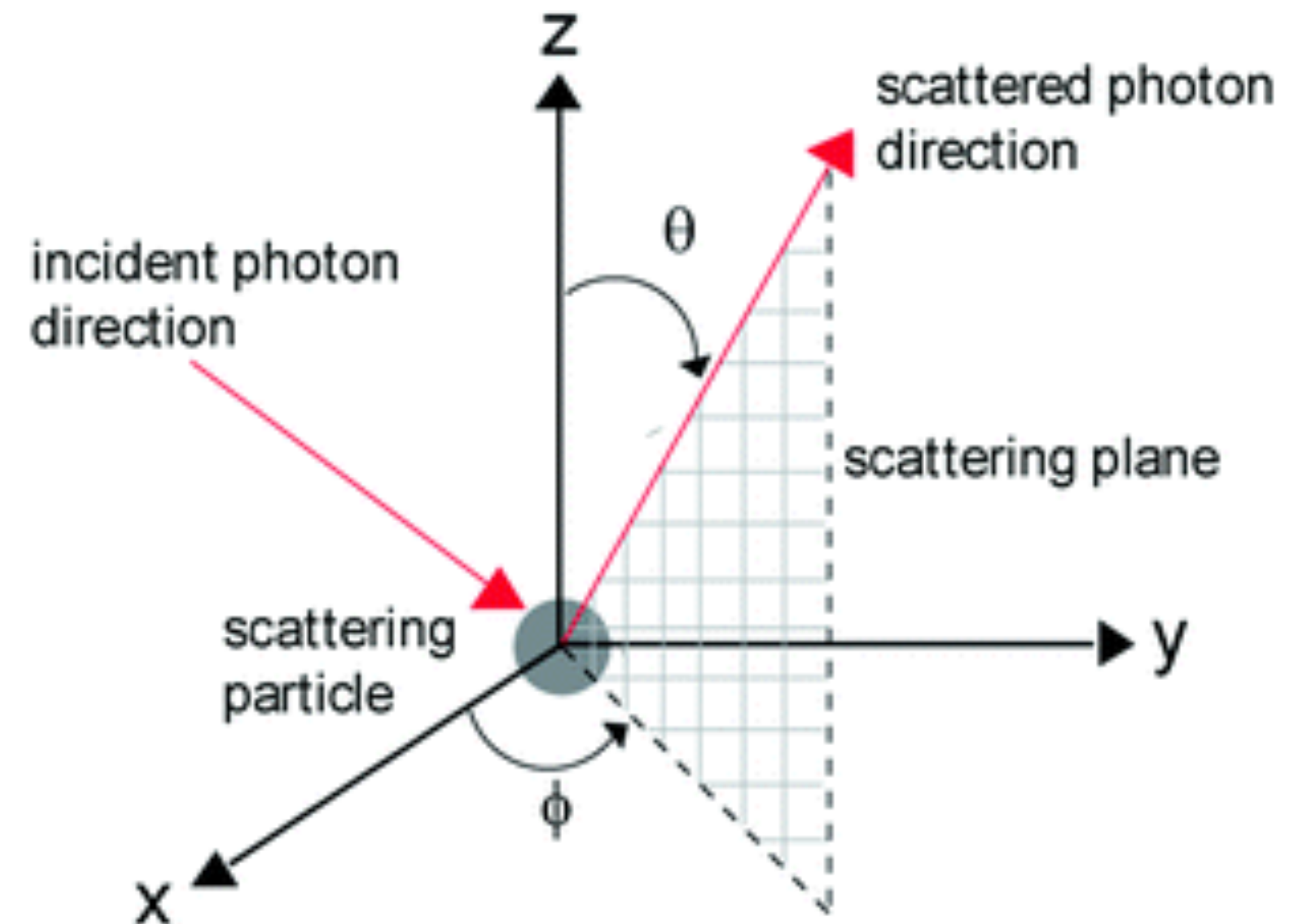
Diffraction integrals:

$$I_m(u, v) = \int_0^\alpha \cos^{1/2} \theta \sin^{m+1} \theta J_m \left(\frac{v \sin \theta}{\sin \alpha} \right) \exp \left(\frac{ju \cos \theta}{\sin^2 \alpha} \right) d\theta$$

Where: $u = \frac{2\pi}{\lambda} \sin^2 \alpha \cdot z$, $v = \frac{2\pi}{\lambda} \sin \alpha \cdot r$

Current Approaches

- **Monte Carlo Methods**
 - Gold standard for: tissue optics,
 - Can handle complex geometries
 - Established for random media



Richards-Wolf Vector Diffraction Theory

Vector Field Components

- When light is focused through a high-NA lens, we can't ignore that light is a vector wave. The electric field has three components (E_x , E_y , E_z), and at high NA, the E_z component becomes significant.

$$E_x(P) = -i(I_0 + I_2 \cos 2\phi)$$

$$E_y(P) = -iI_2 \sin 2\phi$$

$$E_z(P) = -2I_1 \cos \phi$$

Richards-Wolf Vector Diffraction Theory

Diffraction Integrals

- I_0 : Main focusing term, gives Airy pattern
$$I_0(u, v) = \int_0^\alpha \sqrt{\cos \theta} \sin \theta (1 + \cos \theta) J_0 \left(\frac{v \sin \theta}{\sin \alpha} \right) \exp \left(\frac{i u \cos \theta}{\sin^2 \alpha} \right) d\theta$$
- I_1 : Longitudinal field (E_z) — appears at high NA
$$I_1(u, v) = \int_0^\alpha \sqrt{\cos \theta} \sin^2 \theta J_1 \left(\frac{v \sin \theta}{\sin \alpha} \right) \exp \left(\frac{i u \cos \theta}{\sin^2 \alpha} \right) d\theta$$
- I_2 : Cross-polarization correction — breaks circular symmetry
$$I_2(u, v) = \int_0^\alpha \sqrt{\cos \theta} \sin \theta (1 - \cos \theta) J_2 \left(\frac{v \sin \theta}{\sin \alpha} \right) \exp \left(\frac{i u \cos \theta}{\sin^2 \alpha} \right) d\theta$$

Optical Coordinates

- u : Position along optical axis
- v : Position perpendicular to axis
- α : Maximum Half Angle
- **Convenience:** Normalized coordinate—results scale with wavelength and NA
- $(u, v) = (0, 0)$ is the geometric focus
- One set of calculations works for any λ and NA

$$u = \frac{2\pi}{\lambda} \sin^2 \alpha \cdot z$$

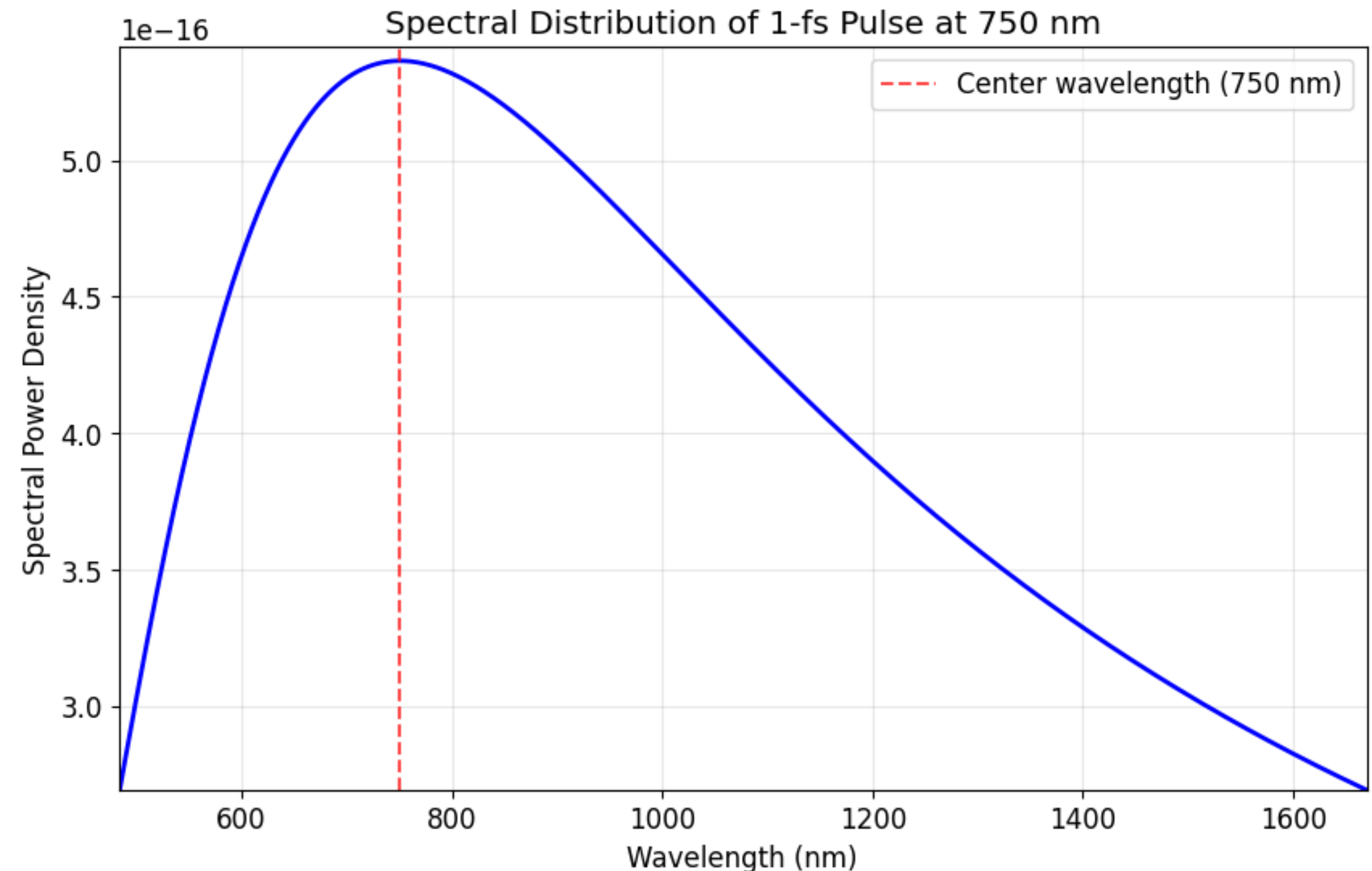
$$v = \frac{2\pi}{\lambda} \sin \alpha \cdot r$$

$$\alpha = \arcsin \left(\frac{\text{NA}}{n} \right)$$

Optical Pulses

Few-Cycle Pulse Characteristics

- Pulse widths: $<10^{-15}$ s (attosecond to femtosecond range)
- Spectral broadening: Bandwidth increases as τ^{-1} (pulse width)
- Critical threshold: $\tau < 1.2$ fs leads to rapid focal quality deterioration
- Coherence length (359.7 nm) $<$ Nyquist sampling (375 nm) for 750 nm, 1.2 fs pulse $\rightarrow$ Information loss through aliasing



Introducing Gaussian Inputs

Incorporating Gaussian Beams into Richards-Wolf

- Romallosa uses uniform illumination (flat intensity across the aperture).
- Real lasers have Gaussian profiles: intensity falls off from center.
- Updated apodization:
- **Truncation Coefficient**
 - $\alpha \rightarrow 0$: Uniform illumination (flat)
 - $\alpha = 1$: Moderately truncated Gaussian
 - $\alpha \geq 4$: Untruncated Gaussian (beam smaller than aperture)

$$A(\theta) = \exp\left(-\alpha \frac{\sin^2 \theta}{\sin^2 \alpha_{\max}}\right)$$

$$I_0(u, v) = \int_0^\alpha \boxed{A(\theta)} \sqrt{\cos \theta} \sin \theta (1 + \cos \theta) J_0\left(\frac{v \sin \theta}{\sin \alpha}\right) \exp\left(\frac{i u \cos \theta}{\sin^2 \alpha}\right) d\theta$$

When does Gaussian illumination behave like uniform?

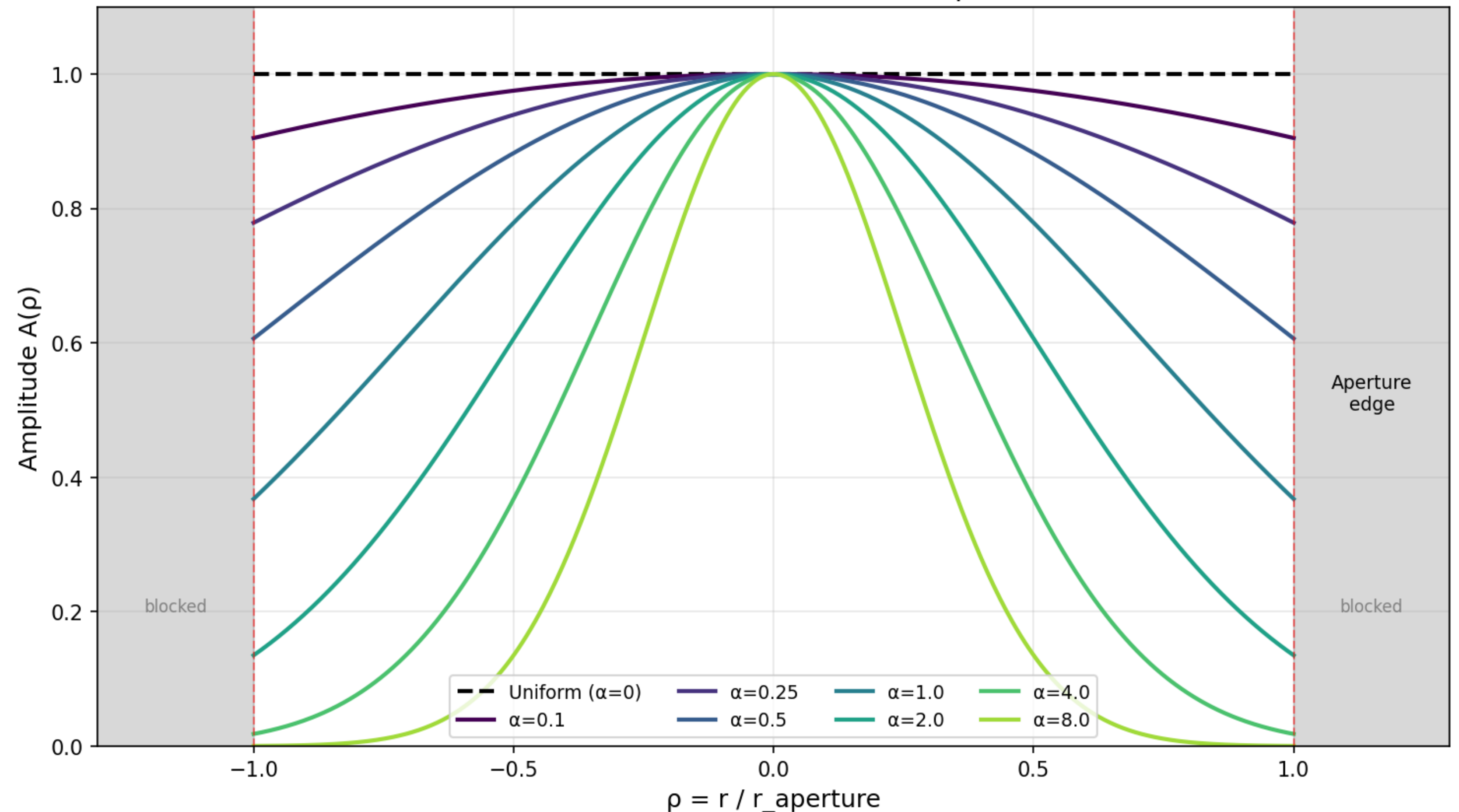
Truncation parameter α :

$$\alpha = \left(\frac{r_{\text{aperture}}}{w_{\text{beam}}} \right)^2$$

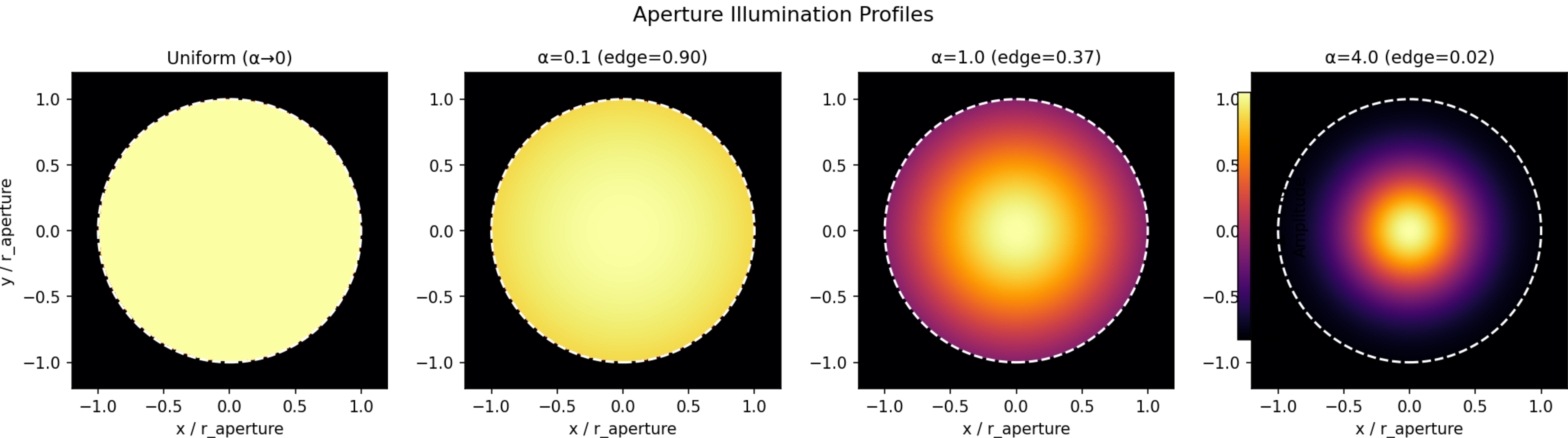
Gaussian amplitude across aperture

$$A(\rho) = \exp(-\alpha\rho^2)$$

- α : truncation coefficient
- r_{aperture} : aperture radius
- w_{beam} : Gaussian beam waist ($1/e^2$ radius)
- $\rho = r / r_{\text{aperture}}$: normalized radial position



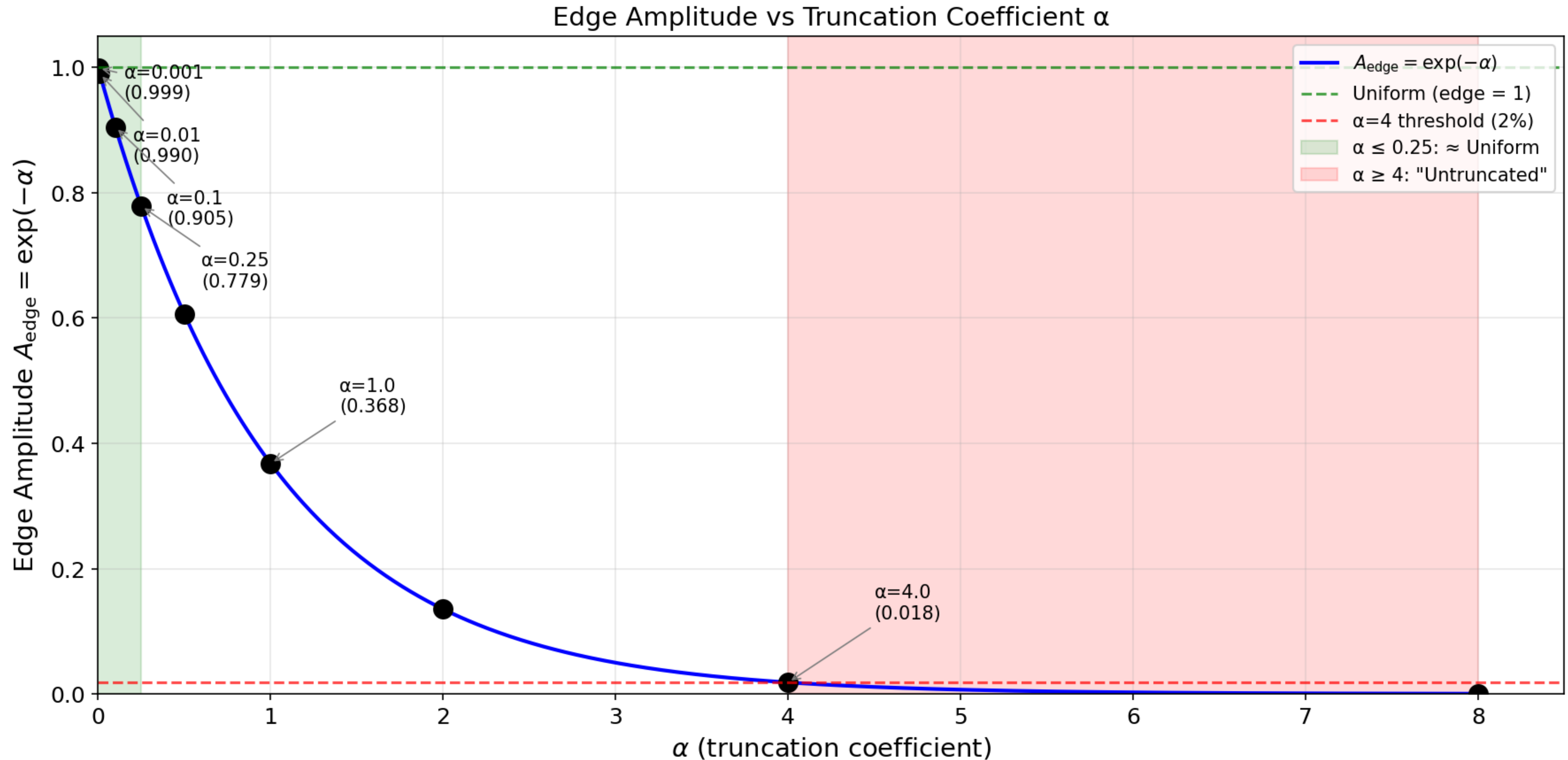
When does Gaussian illumination behave like uniform?



- - $\alpha \rightarrow 0$: beam much wider than aperture $\rightarrow$ uniform illumination
- - $\alpha = 1$: beam waist equals aperture radius $\rightarrow$ 37% at edge
- - $\alpha \geq 4$: beam narrower than aperture $\rightarrow$ "untruncated" Gaussian (2% at edge)

When does Gaussian illumination behave like uniform?

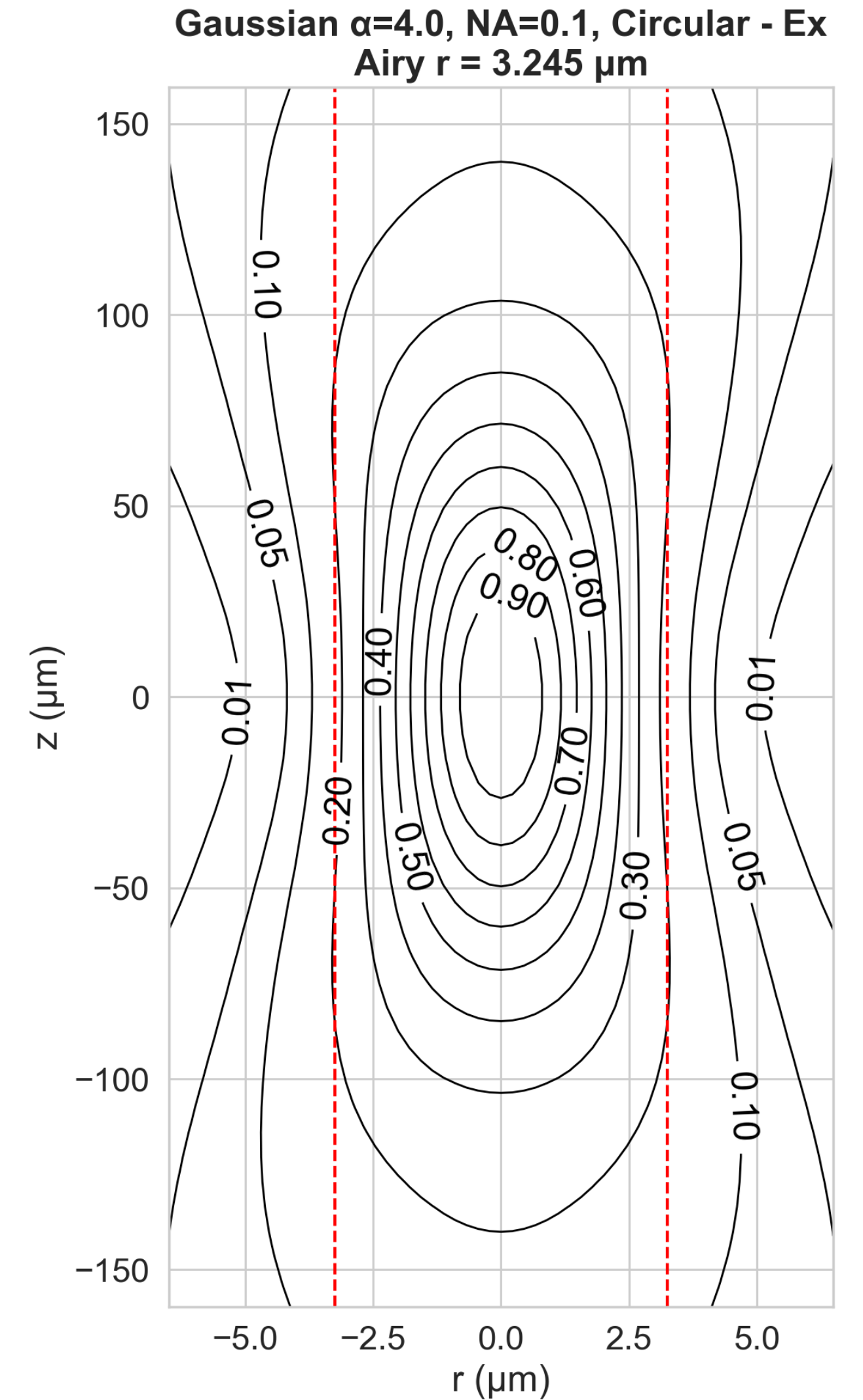
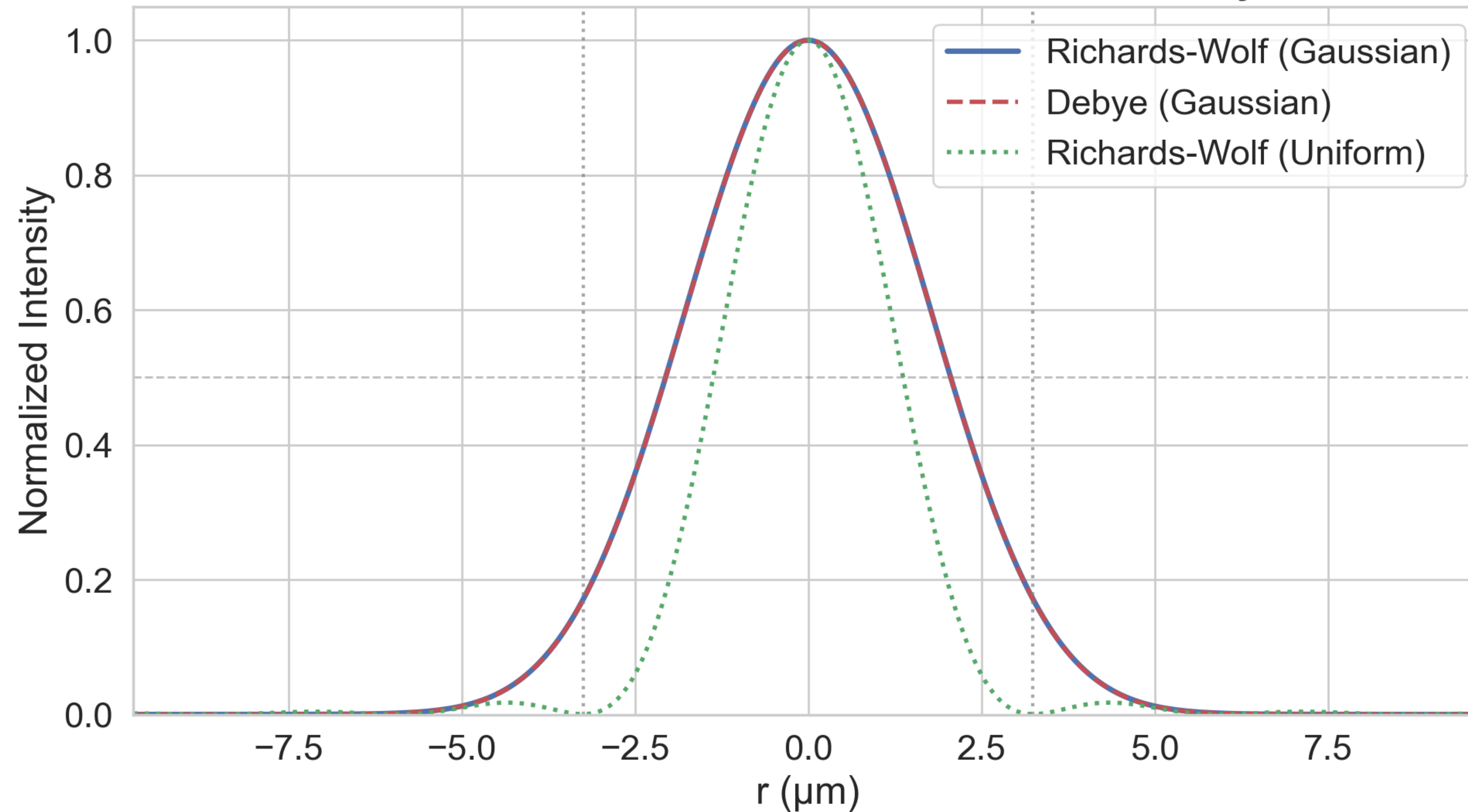
$\alpha \leq 0.01$ is effectively uniform



Focal plane intensity

$\alpha = 4$, $NA = 0.1$

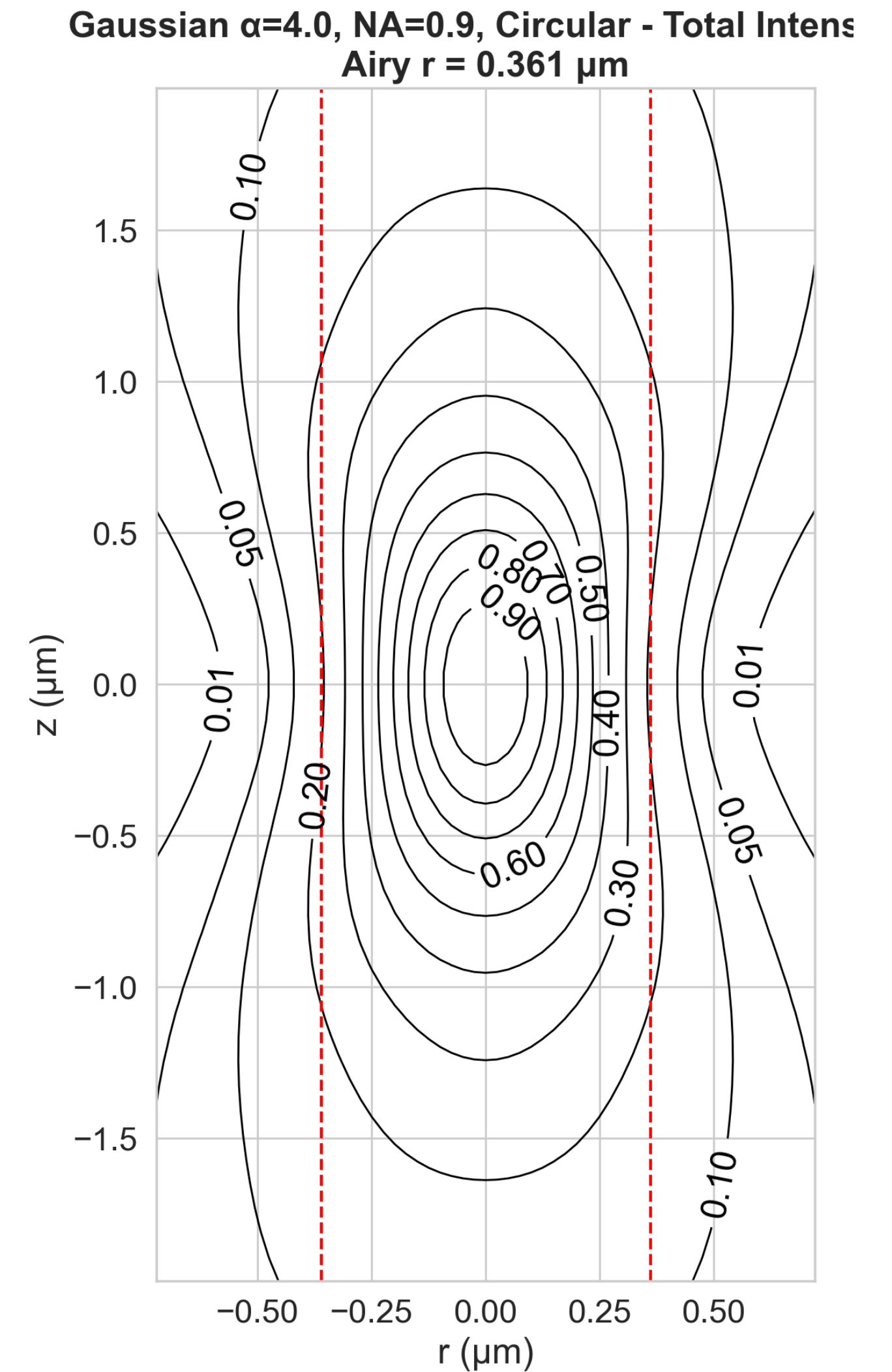
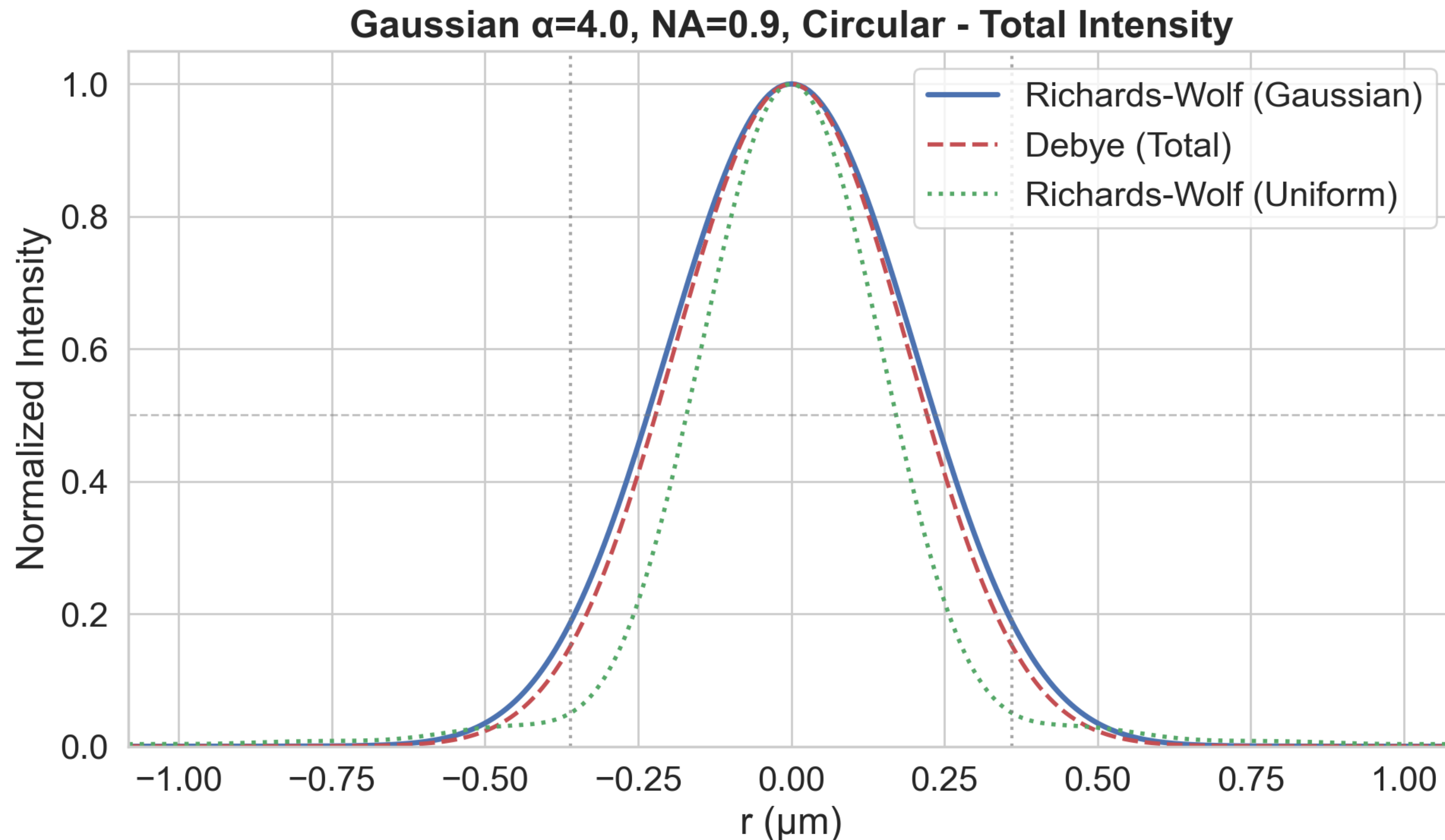
Gaussian $\alpha=4.0$, $NA=0.1$, Circular - Total Intensity



- RW implementation matches Debye analytic results

Focal plane intensity (Uniform)

$\alpha = 4$, $NA = 0.1$, Total Intensity



- Richards-Wolf (Gaussian) starts having additional E_z components

At what NA does scalar (Debye) theory diverge from vector (Richards-Wolf)?

NA Convergence Study

- At what NA does scalar (Debye) theory diverge from vector (Richards-Wolf)?
 - Scalar theory is fast but approximate
 - Vector theory is accurate but slower
 - Need to know when the approximation is acceptable
- Key result: Uniform illumination allows scalar approximation at much higher NA than Gaussian

Comparison Metrics

- **RMSE**

- Unitless (profiles normalized to peak = 1). Measures point-by-point difference.

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(I_{\text{RW}}(r_i) - I_{\text{Debye}}(r_i) \right)^2}$$

- **Pearson Correlation**

- Measures shape similarity. Can be high even when RMSE is large.

$$\rho = \frac{\sum_i (I_{\text{RW},i} - \bar{I}_{\text{RW}})(I_{\text{Debye},i} - \bar{I}_{\text{Debye}})}{\sqrt{\sum_i (I_{\text{RW},i} - \bar{I}_{\text{RW}})^2} \sqrt{\sum_i (I_{\text{Debye},i} - \bar{I}_{\text{Debye}})^2}}$$

- **FWHM Difference**

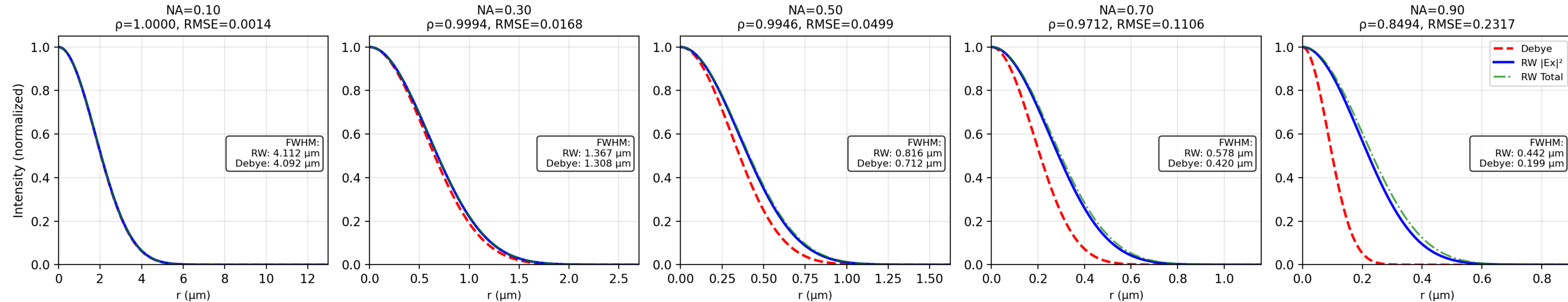
- Spot size prediction error, often what matters most for applications.

$$\Delta\text{FWHM} = \frac{\text{FWHM}_{\text{RW}} - \text{FWHM}_{\text{Debye}}}{\text{FWHM}_{\text{Debye}}} \times 100 \%$$

NA Convergence - Intensity Profiles

Gaussian

Gaussian Illumination ($\alpha=4$): Intensity Profiles

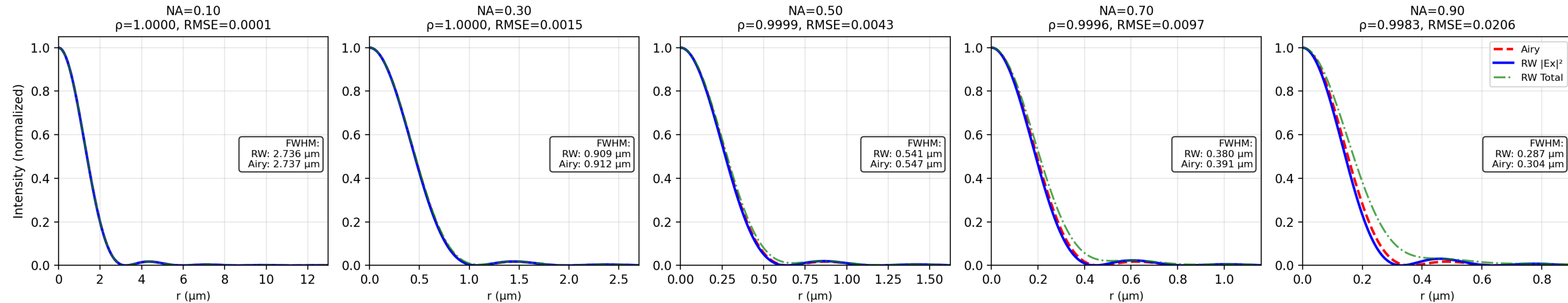


- At low NA (0.1, 0.3): profiles nearly identical
- At high NA (0.7, 0.9): clear divergence, RW is broader

NA Convergence - Intensity Profiles

Uniform

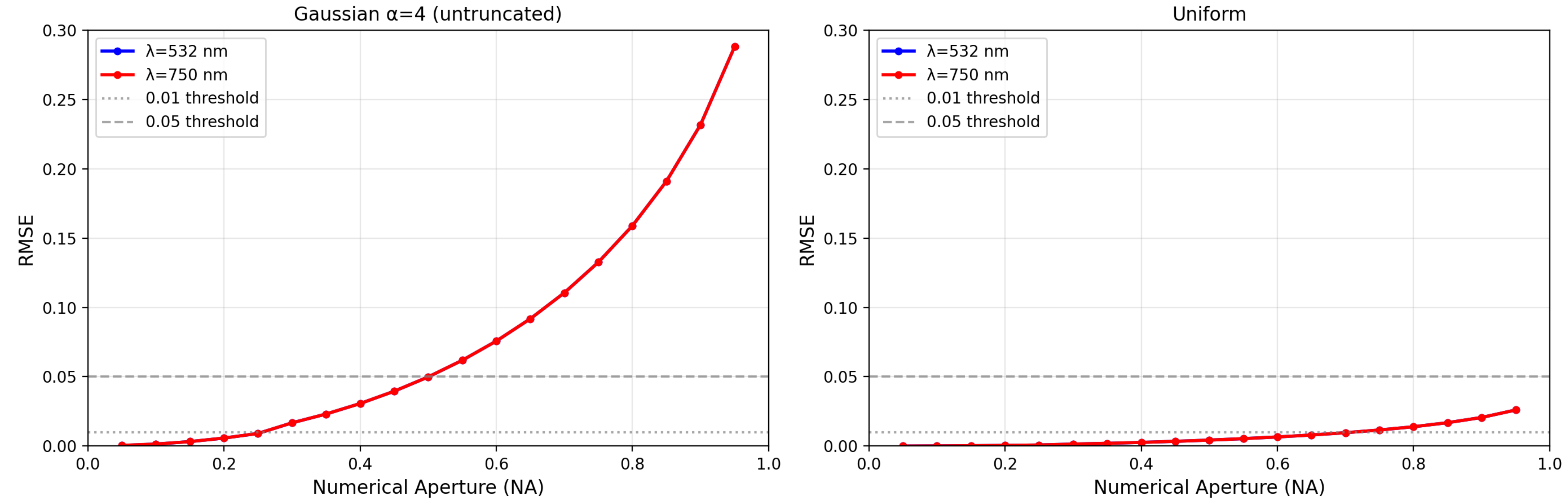
Uniform Illumination: Intensity Profiles



- Slower divergence compared to Gaussian input

NA Convergence - RMSE

Richards-Wolf vs Debye: RMSE vs NA

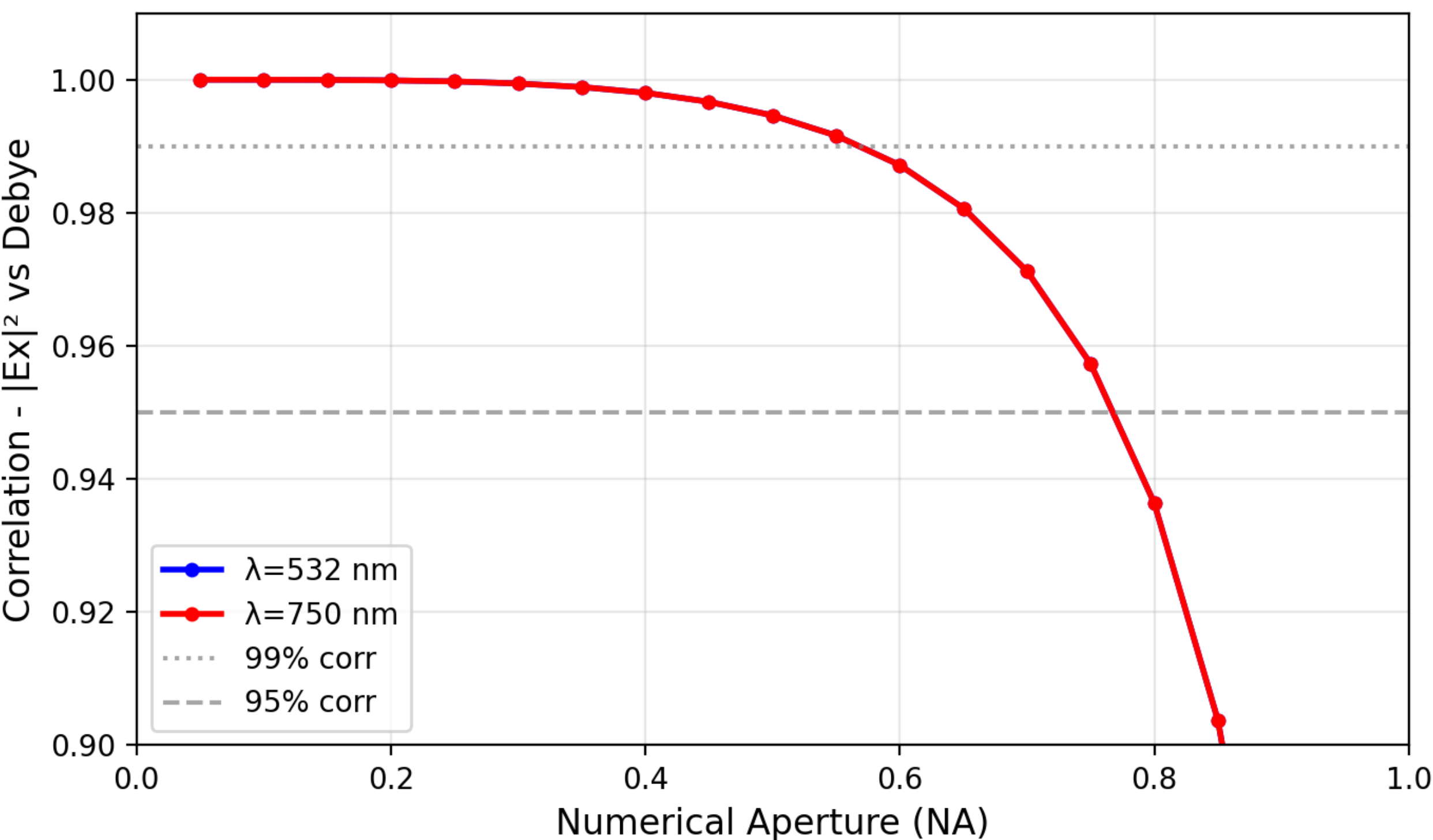


- Gaussian $\alpha=4$: RMSE > 0.01 at NA ≈ 0.30 , RMSE > 0.05 at NA ≈ 0.55
- Uniform: RMSE > 0.01 at NA ≈ 0.75 , RMSE > 0.05 at NA ≈ 0.95
- Wavelength doesn't matter: same thresholds for 532 nm and 750 nm
- Uniform illumination allows scalar approximation at much higher NA

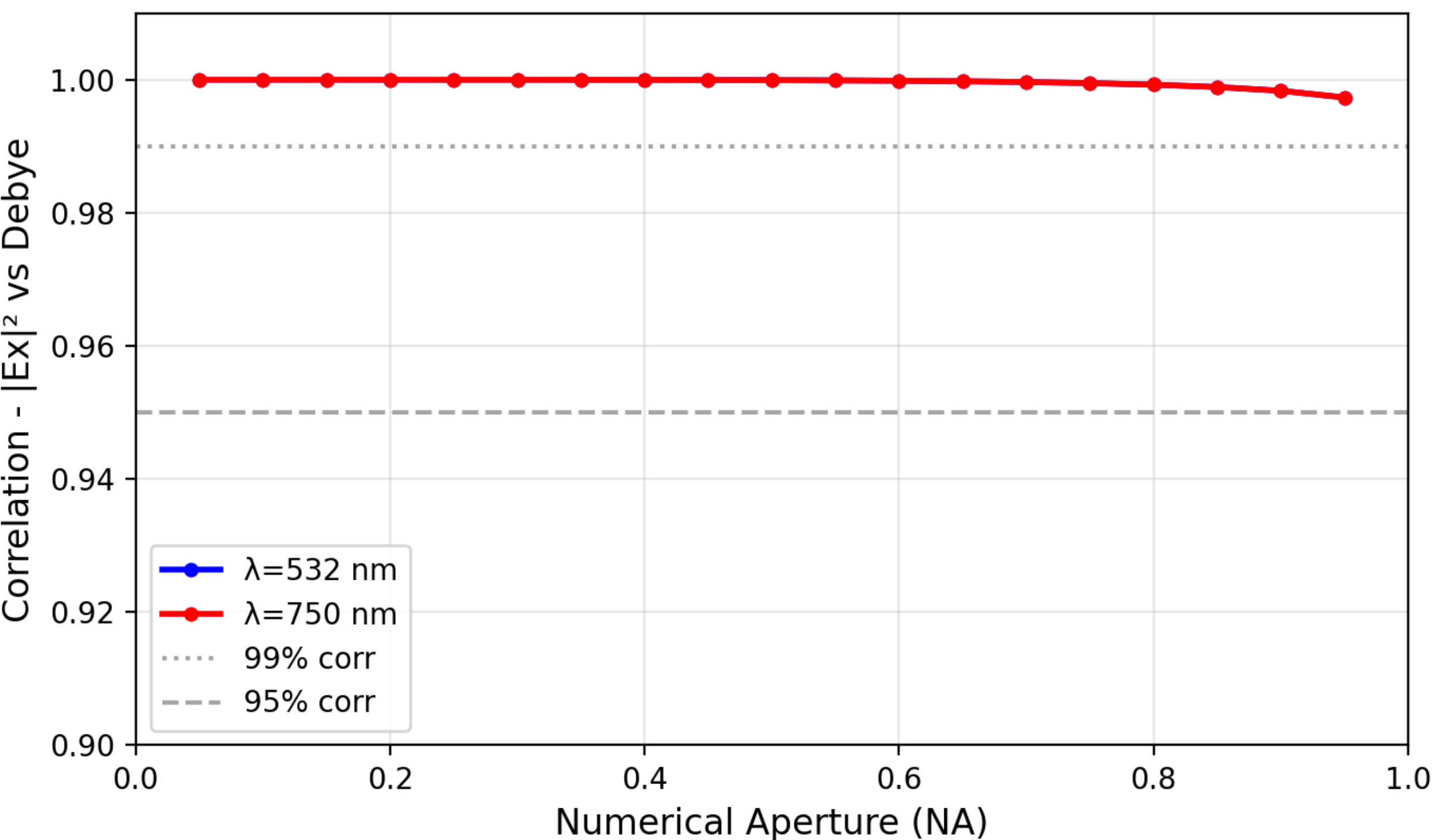
NA Convergence - Correlation

Richards-Wolf vs Debye: Correlation vs NA
Comparison across wavelengths

Gaussian $\alpha=4$ (untruncated)

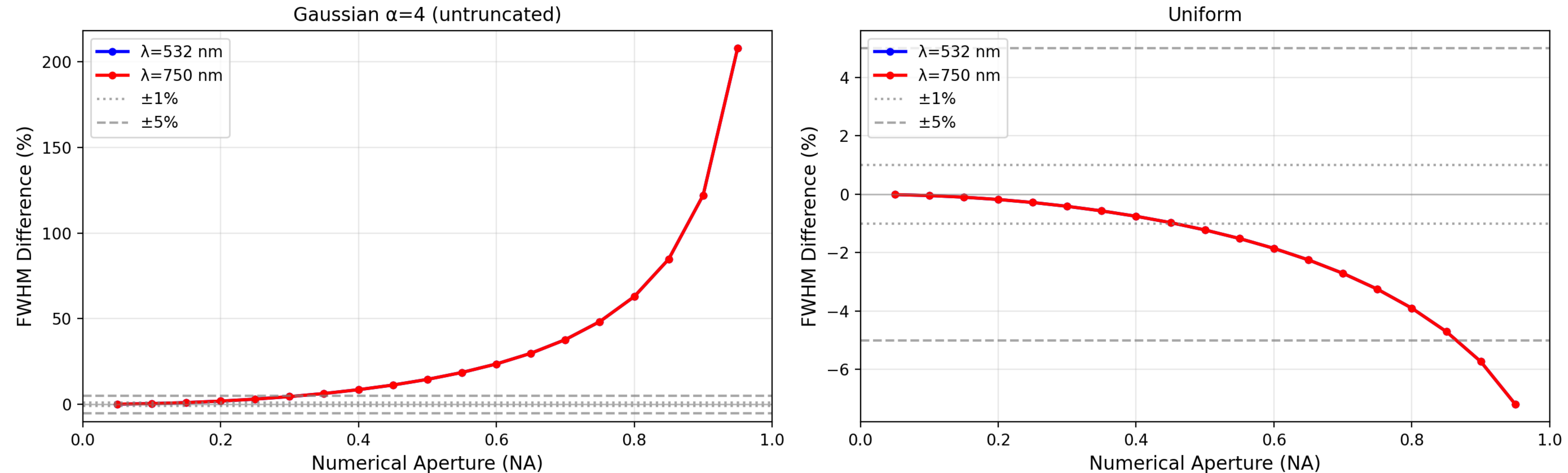


Uniform



NA Convergence - FWHM Difference

Richards-Wolf vs Debye: FWHM Difference vs NA
(RW - Debye) / Debye $\times$ 100%



- Gaussian: FWHM diverges faster
- Uniform: FWHM stays within $\pm 7\%$ even at NA=0.95

Extended DOF - Annular Apertures

Extended DOF — Annular Apertures

Babinet's Principle:

$$E_{\text{annular}}(\varepsilon) = E_{\text{disk}}(\text{NA}_{\text{outer}}) - E_{\text{disk}}(\text{NA}_{\text{inner}})$$

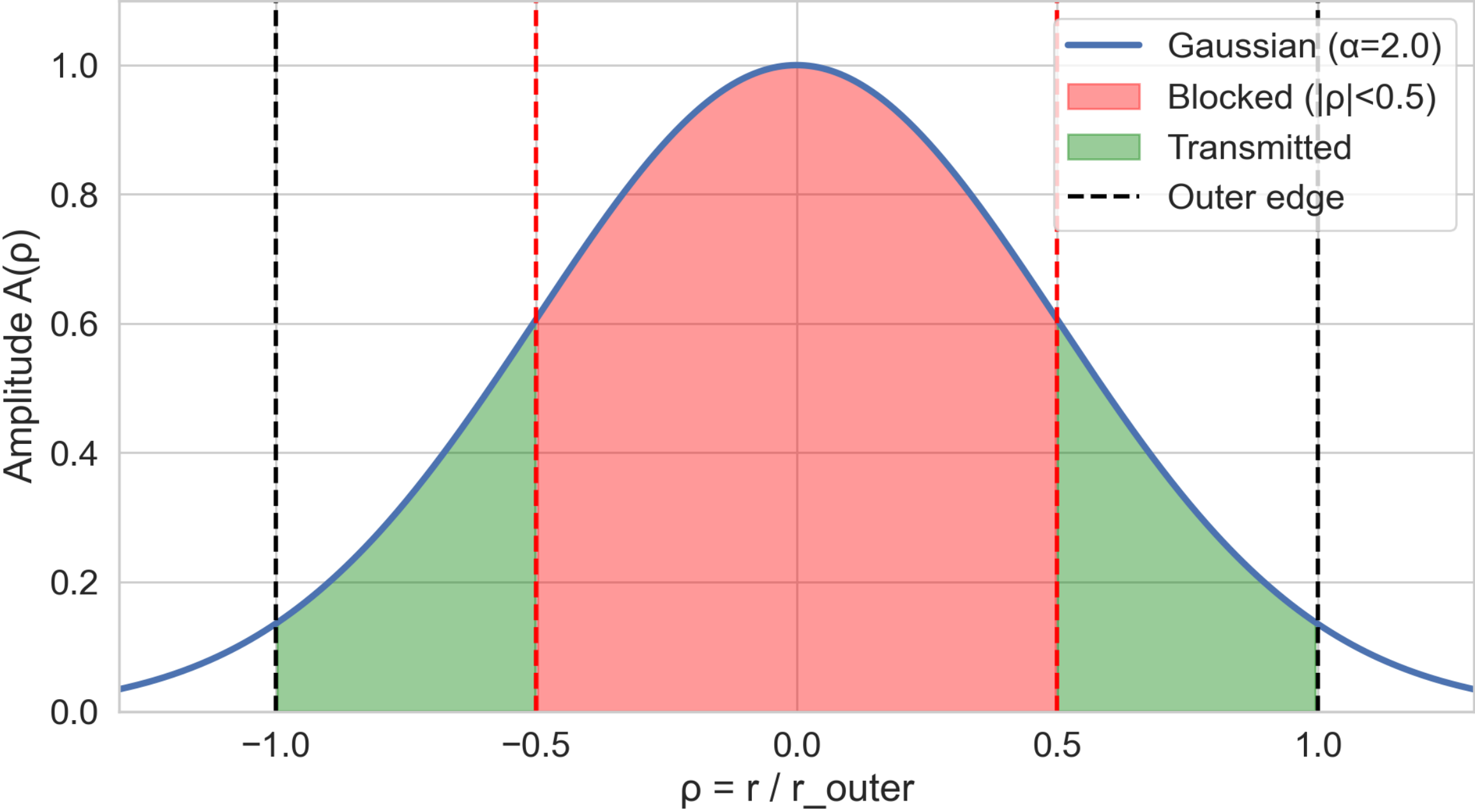
- Applied Babinet's principle to model annular apertures
 - Higher obstruction ratio (ε) $\rightarrow$ narrower central peak, improved resolution, stronger side lobes

$$\varepsilon = \frac{r_{\text{inner}}}{r_{\text{outer}}}$$

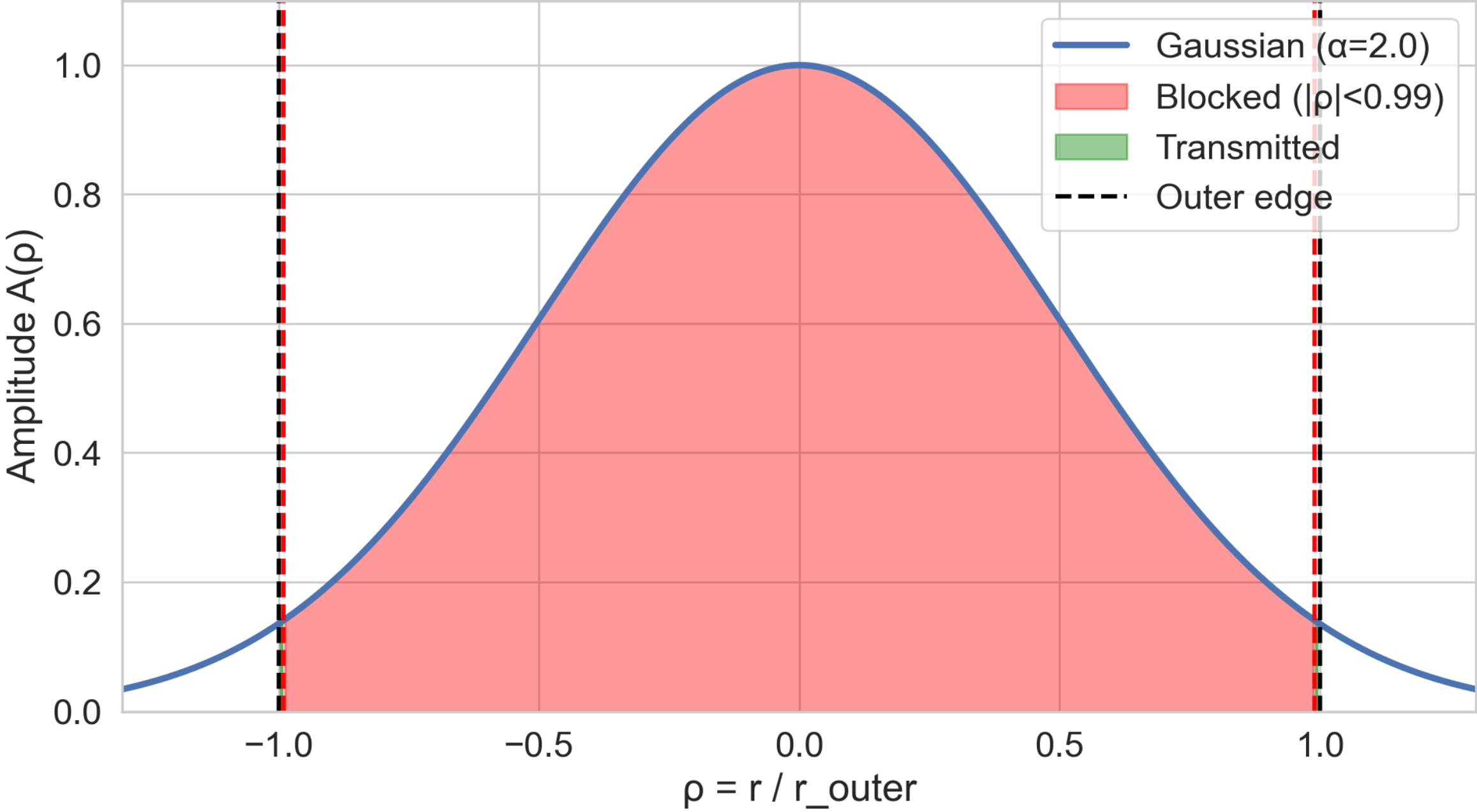
$$\text{DOF}(\varepsilon) = \frac{\text{DOF}_0}{1 - \varepsilon^2}$$

Gaussian Truncation

Gaussian on Annular Aperture ($\epsilon = 0.5$)

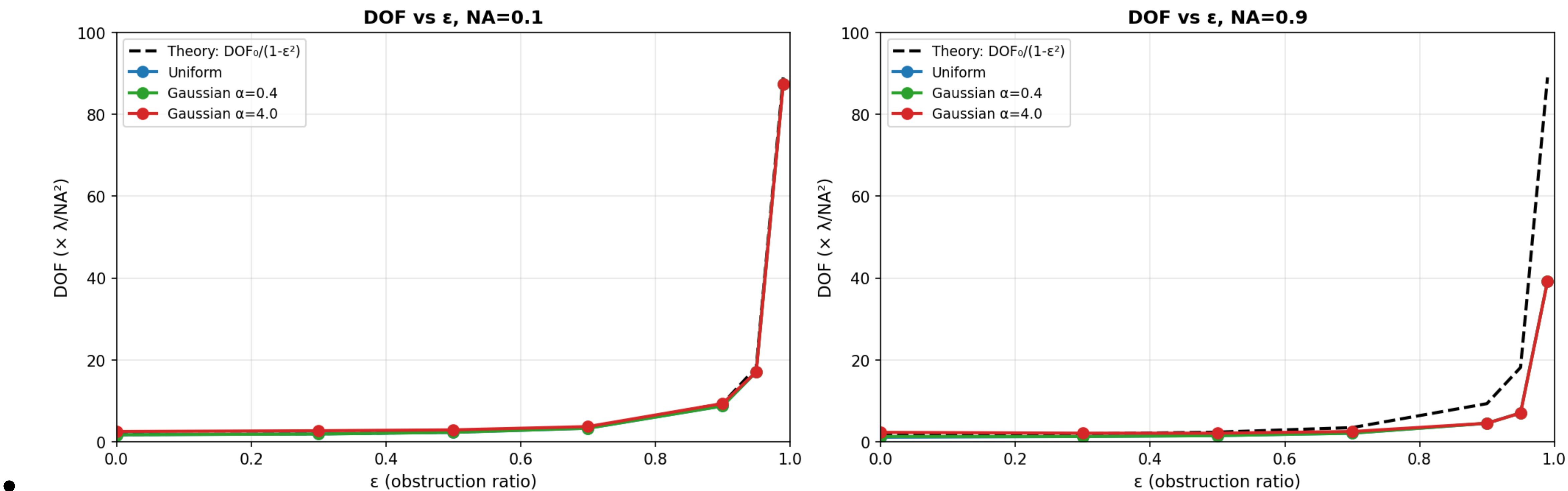


Gaussian on Annular Aperture ($\epsilon = 0.99$)



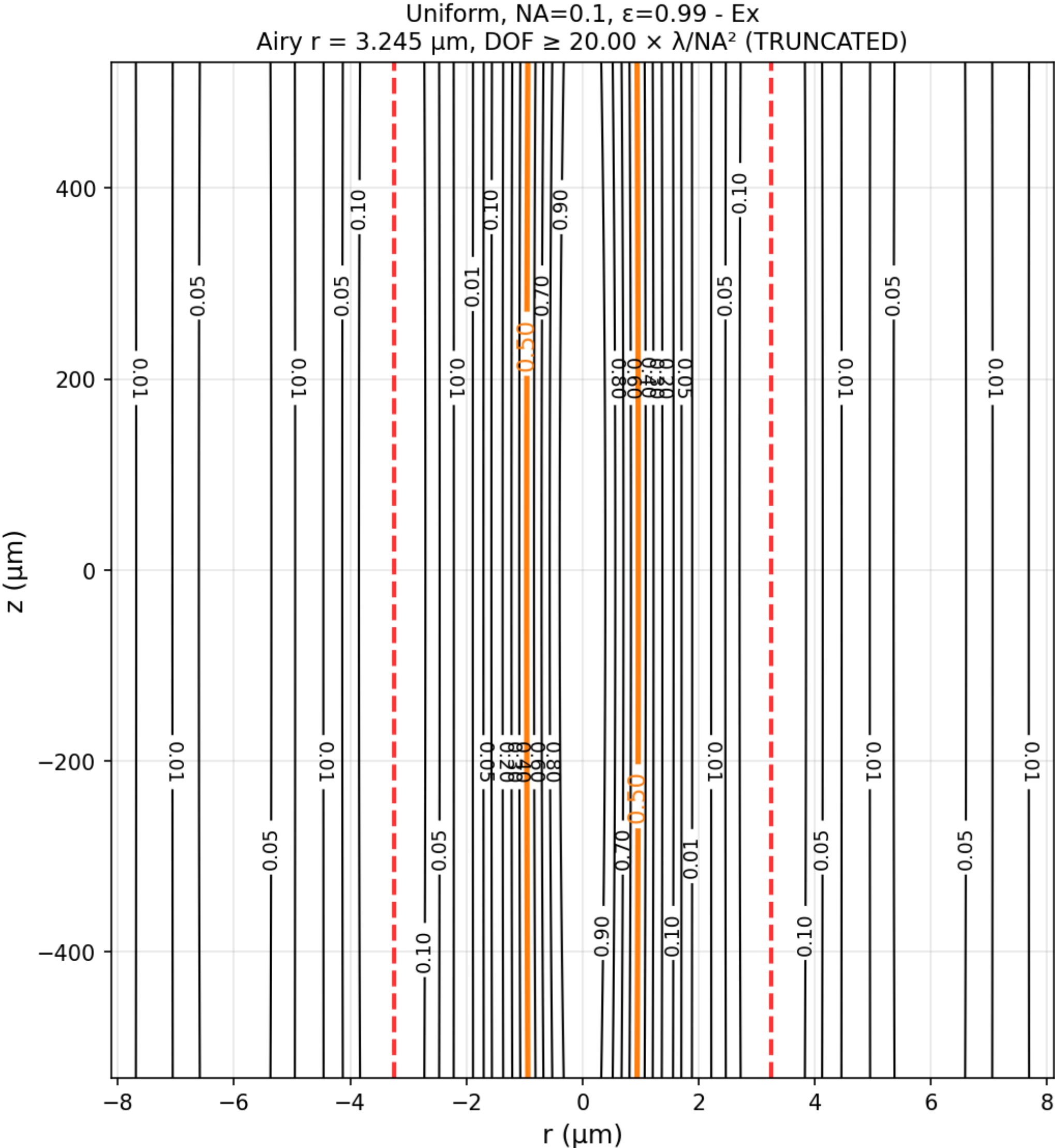
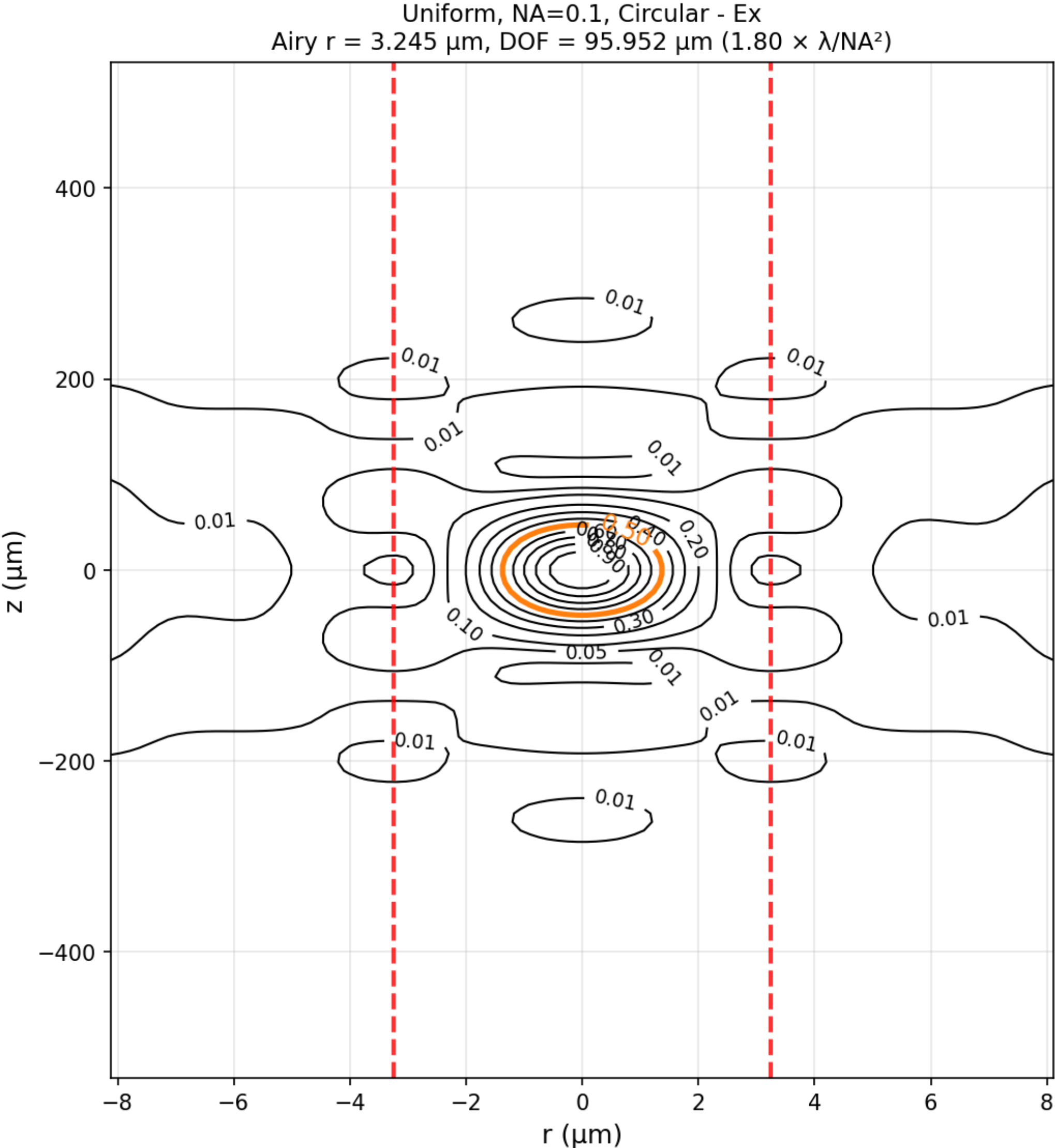
Effect of Increasing Obstruction Ratio on DOF

DOF vs Obstruction Ratio: Comparing Illumination Types



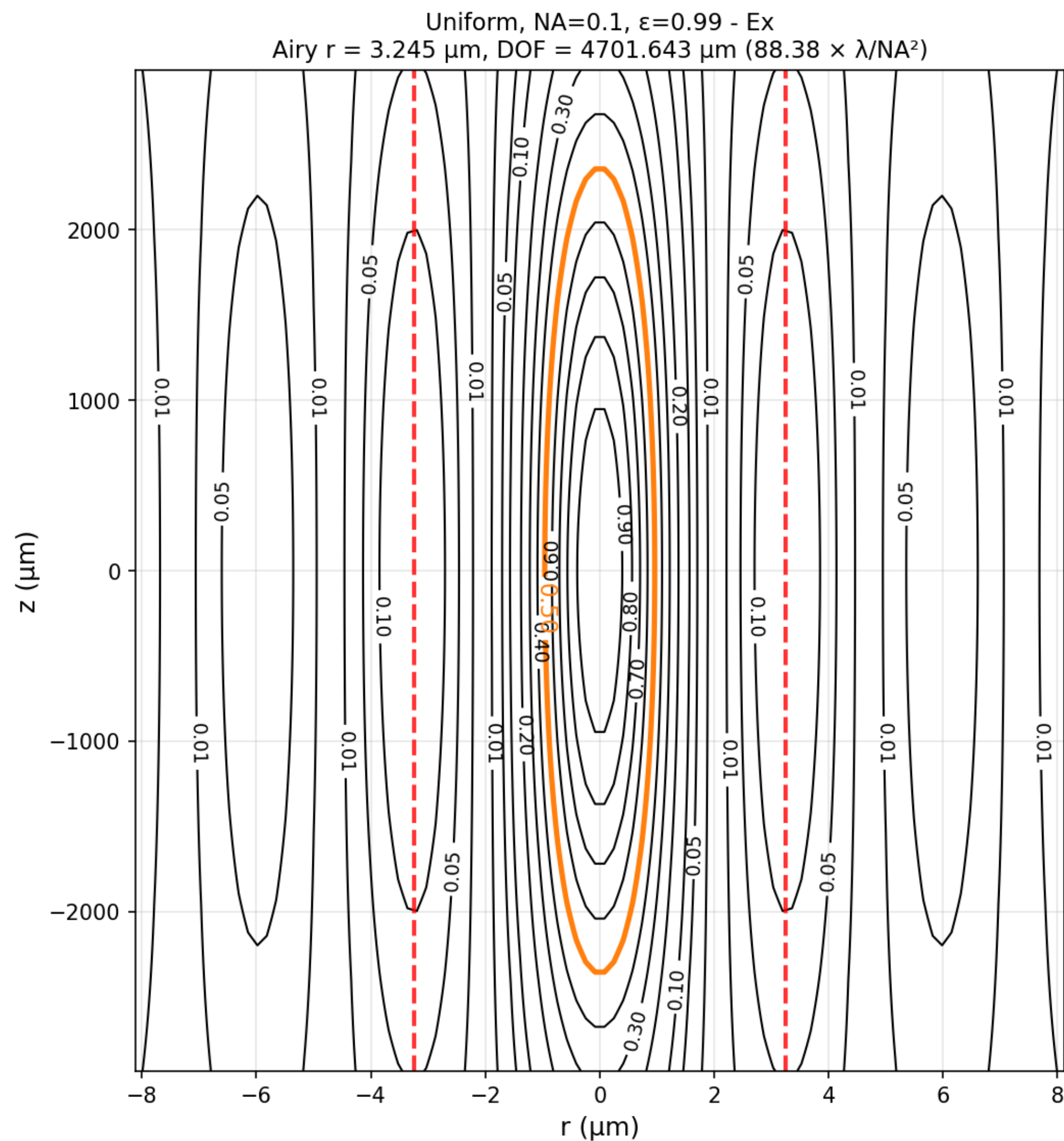
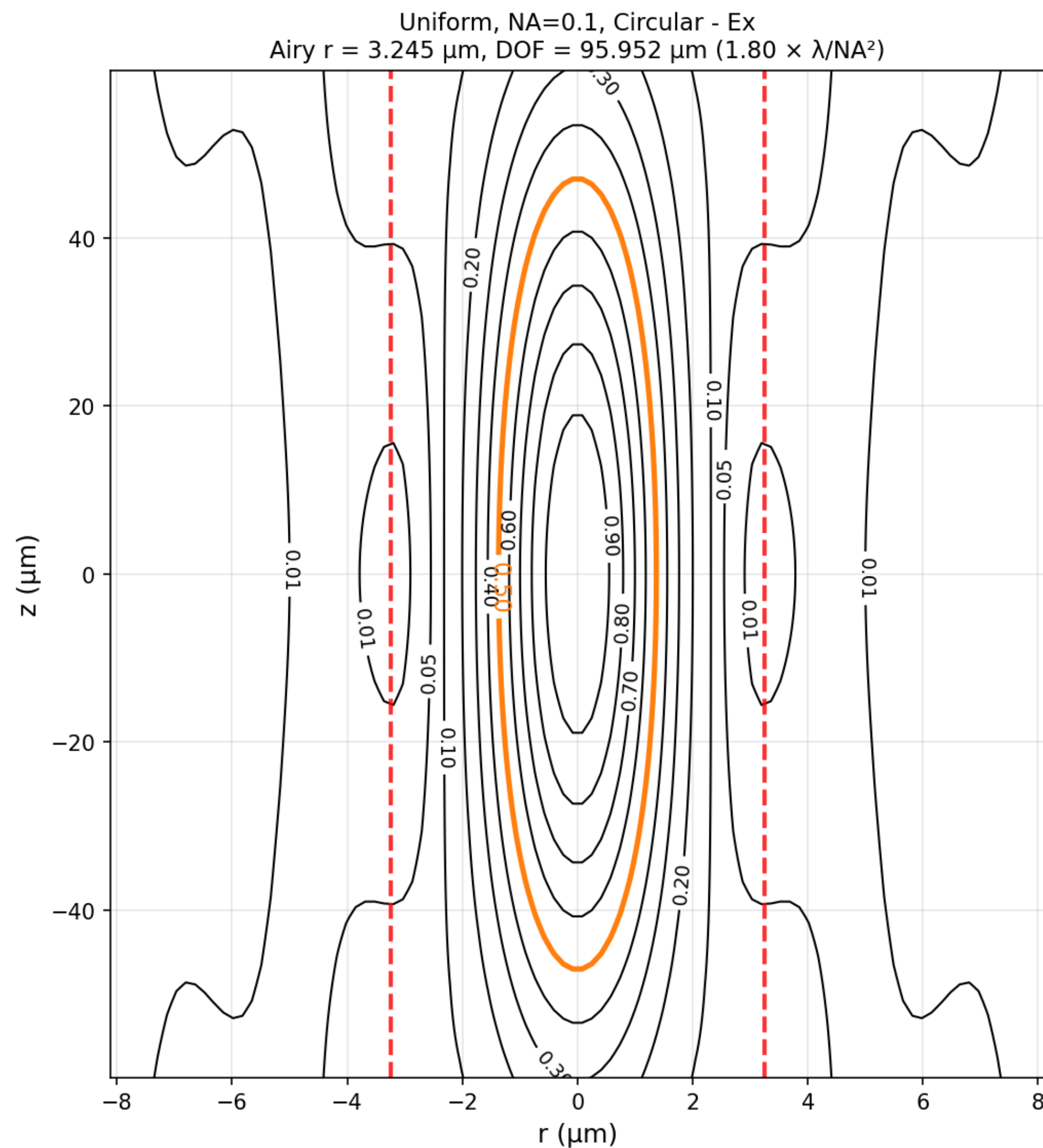
Extended DOF

Low NA



Extended DOF

Low NA



Summary

1. Reproduced Romallosa 2003 figures for high-NA optical pulses
2. Introduced Gaussian apodization factor for realistic beam profiles
 - Validated convergence to Debye/Airy theory at low NA
3. Quantified NA thresholds where vector diffraction becomes necessary
4. Implemented annular apertures via Babinet's principle
5. Characterized DOF extension vs obstruction ratio for annular apertures

Compared uniform vs Gaussian illumination across all configurations

Next Steps

- **Immediate:** Optical Pulses + Annular Apertures
 - Implement spectral integration for pulsed sources
 - Study how pulse duration interacts with annular DOF extension
 - Key question: Does extended DOF help or hurt for pulsed sources?
- **Future:** Apply Monte Carlo with scattering to systems above

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Progress Report

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